PROCESSING GUIDELINES

Laminate: Synamic 6
Prepreg: Synamic 6B

Very Low Loss & Heat Resistance Multilayer Material

This product uses IPC-4101E Standards as a reference, and Shengyi make some changes according to the product characteristics of the actual situation as to making it more suitable for Shengyi Synamic6/Synamic6B product use.

1. Storage Condition

1.1 Laminate core

1.1.1 Store Condition

- Pack with original forms on the platform or on the appropriate frame, avoiding stress, prevent sheet deformation caused by inappropriate storage which may impact the subsequent PCB processes.

1.1.2 Storage Environment

- Sheets should be stored in ventilated, dry, at room temperature under environment control, avoiding direct sunlight, rain and avoid erosion of corrosive gas (stored environment directly affect the quality of material).
- For double-sided copper-clad boards (cores), to minimize shifting as to avoid scratching the surface of the product, with a suitable environment and condition for storage, the shelf life can be up to two years.

1.1.3 Operation Manuel

- Wear clean gloves and carefully operate the cores. Copper foil collisions, sliding will cause damage of the cores.
- Bare hands action will cause contamination to copper foil surface. These defects are likely to cause adverse effects.

1.2 Prepreg

1.2.1 Storage Condition

- Levels stored in original packaging form, avoiding stress, prevent sheet deformation caused by inappropriate storage condition.
- Leftover or cut Prepregs should pack and seal with vacuum foil packaging and put it back in the original packaging tray.

1.2.2 Storage Environment

- Prepreg sealed packaging should be stored in free of UV irradiation environment, specific storage conditions and the storage period as follows:

- Condition 1. At a temperature of <5℃, storage period for 6 months.
- Condition 2. At a temperature of <23℃, relative humidity <50% when stored, storage period for 3 months

- Note: Relative humidity affect prepreg quality the most, pay special attention on weather (conduct dehumidification process is necessary for wet weather).

2. Cutting Guideline

- Cutting the best way is left to professional staff wear clean gloves during operation, prevent the pollution of prepreg surface; operation must be careful to prevent prepreg wrinkle or crack, to avoid affect prepregs.

3. Prepregs Use Recommendations

- If moving from a low temperature storage space to a higher temperature or ambient temperature storage space, it must go through the temperature settle process, (8 - 24 hours, settle time is varies depending on temperature variation in between two storage conditions). Open package after temperature settle process is completed as to avoid affecting the quality and adhesion of prepregs.
- For PP package stored in above conditions 1 or 2, after open is required to complete the use as soon as possible, for packages opened more than 3 day, it must re-inspect and insure quality before use.
- Leftover or cut prepregs should pack and seal with vacuum foil packaging and put it back in the above stated storage condition 1 or 2.

4. PCB Process Recommendations

4.1 Baking on the cores should be carried out before use to eliminate internal stress. Baking conditions should be 150~160℃/4~6 hours or 170~180℃/2~4 hours;

4.2 After Brown Oxide treatment, it is recommended to bake 120℃/60-120min as to remove any surface moisture before press lamination process. Material should be used within 4 hours after baking.

4.3 Avoid turn over actions in prepreg and core stacking, to reduce possible deformation.

4.4 Press Recommendations

4.5.3 Vacuum press.

4.5.4 Press lamination suggested heating rate 2.5~4℃/min (suggested 3.0~3.5℃ /min, material temperatures 80~140℃ in the region.

4.5.5 Lamination pressure setting, outer material temperature at about 80~100℃ when increasing pressure (Note: The full pressure to use is 400-500PSI, suggested 450PSI).

4.5.6 Curing material temperature above 210℃, and keep at this temperature at above 120min. After curing, cooling rate should be less than 2.5℃/min, material's temperature for die open should be less than 130℃.

4.5.7 If used dummy layer without copper foil or single sided board in multilayer lamination, please roughen the surface of the dummy board or single-sided board, to avoid weak adhesive in bonding strength.

4.5 Drilling

4.5.1 New drilling bit is recommended to use when drilling. To ensure a good quality of hole wall. The life of new drills and drilling head should be limited to reduced life of 300~1000 hits (suggested 500 hits).

4.5.2 After drilling, it's recommend to bake for 3-4 hours at 190℃. Avoid direct contact with heat.

4.5.3 Laser drilling is acceptable, whose specific parameters should be adjusted according to different devices.

4.5.4 LE aluminum or bakelite cushion board is recommended to help.

Table 1: Drilling parameters (reference only)

Diameter		Infeed	Spindle Speed	Retract Rate	Hit counts	Chip load
mm	inch	ipm	krpm	ipm	H	mil/rev
0.20	0.0078	67	95	500	500	0.71
0.25	0.0097	71	95	500	500	0.75
0.30	0.0117	76	95	500	500	0.80
0.35	0.0136	81	95	500	500	0.85
0.40	0.0156	105	63	500	500	1.67
0.45	0.0175	110	60	500	500	1.83
0.50	0.0195	125	62	1000	500	2.02
0.55	0.0214	127	60	1000	500	2.12

0.60	0.0233	130	58	1000	500	2.24
0.65	0.0253	125	55	1000	500	2.27
0.70	0.0272	125	53	1000	500	2.36
0.75	0.0292	125	51	1000	500	2.45
0.80	0.0311	125	50	1000	500	2.50
0.85	0.0331	124	49	1000	500	2.53
0.90	0.0350	123	48	1000	500	2.56
0.95	0.0370	122	47	1000	500	2.60
1.00	0.0389	121	46	1000	500	2.63
1.05	0.0409	121	45	1000	500	2.69
1.10	0.0428	120	44	1000	500	2.73
1.30	0.0506	120	35	1000	500	3.43
1.50	0.0584	115	32	1000	500	3.59
1.70	0.0661	100	28	1000	500	3.57
1.90	0.0739	90	23	1000	500	3.91

Note: Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, stack count and stack height etc.

4.6 Desmear

- Desmear rate of the material is smaller than most of the existing lead-free compatible FR-4 materials. To achieve a better desmearing, it is recommended to use the following conditions:
- Use plasma at first → followed with ultrasonic rinse → add chemical desmear after.
- To guarantee the desmear effect, the total demear rate is recommended to be controlled around 0.4mg/cm^2. Plasma time is better not too long. Below parameter is for your reference.

Table 2: Plasma parameters (Refence only)

	Gas Flow Rate (L/min)			Watts	Time	Temp.	Flow	Pressure
Parameter	O_2	N_2	CF_4	(V)	(min)	(˚C)	Rate (SLM)	(mTorr)
Seg 1	2.25	0.25	0.00	9000	45.0	80.0	2.50	250
Seg 2	2.46	0.24	0.30	6000	15.0	105.0	3.00	220
Seg 3	2.50	0.00	0.00	5000	5.0	100.0	2.50	250

4.7 Punching is not suitable for PCB profiling process, routing is recommended with reduced routing speed. Try to avoid sudden vibration during routing process which may cause board edge detonation.

4.8 Baking is recommended before packing, condition 125℃/4~6h.

4.9 Vacuum sealed with aluminum plastic pack.

5. Soldering

5.1 For finished PCBs, basing on the differences of PCB constructions, the packing and storage condition, the shelf life should change accordingly. Normal shelf life should be finishing assembly within 3 months. After expiration of shelf life, baking is recommended and the condition should be 125℃/4-6 hours.

5.2 Applicable for regular lead-free soldering.

6. Design Recommendations

6.1 Due to Fiberglass structure and weft density differences, when manufacturing it's recommended to use symmetrical PCB construction stackup.

6.2 Selected dielectrics and to the adjacent cores must be compatible in bonding, weft and grain directions, as to avoid warpage and deformations.

6.3 A different resin system with a large variation in resin density, therefore, even with the same resin content (RC) if a different resin, the dielectric thickness will be varies. A specific resin content (RC) for an actual dielectric thickness should consult with Technical Service Engineer.

During using the Shengyi Synamic6/Synamic6B material, if you have any questions and

suggestions, please feel free to contact Shengyi, and Shengyi will provide you with an efficient and effective technical services.